nature genetics

Reference component analysis of single-cell transcriptomes elucidates cellular heterogeneity in human colorectal tumors

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Intratumoral heterogeneity is a major obstacle to cancer treatment and a significant confounding factor in bulk-tumor profiling. We performed an unbiased analysis of transcriptional heterogeneity in colorectal tumors and their microenvironments using single-cell RNA–seq from 11 primary colorectal tumors and matched normal mucosa. To robustly cluster single-cell transcriptomes, we developed reference component analysis (RCA), an algorithm that substantially improves clustering accuracy. Using RCA, we identified two distinct subtypes of cancer-associated fibroblasts (CAFs). Additionally, epithelial–mesenchymal transition (EMT)-related genes were found to be upregulated only in the CAF subpopulation of tumor samples. Notably, colorectal tumors previously assigned to a single subtype on the basis of bulk transcriptomics could be divided into subgroups with divergent survival probability by using single-cell signatures, thus underscoring the prognostic value of our approach. Overall, our results demonstrate that unbiased single-cell RNA–seq profiling of tumor and matched normal samples provides a unique opportunity to characterize aberrant cell states within a tumor.

Intratumoral heterogeneity is a key determinant of tumor biology, treatment response and patient survival[1,2]. It is therefore essential to comprehensively characterize the phenotypes and interactions of the diverse cell types present within tumors. However, traditional molecular profiling studies have largely relied on bulk-tissue analysis, which obscures the signatures of distinct cell populations. One strategy for characterizing intratumoral heterogeneity is to purify cells on the basis of predefined markers[3]. This approach is limited by lack of knowledge of the precise cell types and states that contribute to tumorigenesis, metastasis and drug resistance and is constrained practically by the availability of specific, high-affinity antibodies for purification of all relevant cell populations.

Single-cell sequencing technologies enable comprehensive, unbiased analysis of cellular diversity within tumor masses. Multiple studies have used single-cell genome sequencing to investigate the clonal evolution of tumor cell lineages, measure somatic mutation rates and gain insights into chemotherapeutic drug response[4]. In contrast, transcriptome-wide analyses of cell heterogeneity are not yet as well established. A recent pioneering study of glioblastoma used single-cell RNA sequencing (scRNA–seq) to uncover unexpected variability in stemness, oncogenic signaling and immune response across cells sorted from five tumors[5]. One limitation was that the study was restricted to cancer cells and thus could not examine the tumor microenvironment. A broader subsequent study examined the diversity of functional states occupied by multiple cell types within melanoma tumors but did not include comparison to matched normal samples[6].

Although multiple algorithms have been developed to infer cell type by clustering scRNA–seq profiles[7–11], robust cell type inference remains challenging because of high levels of noise, technical variability and batch effects in data[12,13]. In the case of clinical samples collected under diverse conditions and in multiple batches, this problem is particularly severe. We therefore developed a novel clustering method named reference component analysis (RCA) and compared it to existing algorithms using a novel benchmark data set.

Colorectal cancer (CRC) is the third most common cancer globally[14]. Stromal cell signatures from bulk transcriptomes have recently been shown to correlate with risk of recurrence and predict patient

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Received 1 November 2016; accepted 24 February 2017; published online 20 March 2017; doi:10.1038/ng.3818

survival, highlighting the importance of diverse cell populations in CRC[15–17]. Here we present scRNA–seq analysis of 590 cells (after quality control) isolated without filtering for cell type markers from 11 primary CRC tumors and matched normal samples. Cell type clusters identified by RCA in colorectal scRNA–seq data included epithelial cells, fibroblasts, endothelial cells and major immune cell populations, some of which could be further divided into novel subtypes. We defined the expression signatures of these cell types, as well as their utility in stratifying patients by survival. The scRNA–seq data provided insights into several functional properties of tumor cells, such as pathway activation, stemness and EMT.

RESULTS

RCA outperforms existing algorithms for clustering single-cell transcriptomes

Accurate clustering of single-cell transcriptomes, with each cluster defining a distinct cell type, is required for cell type identification from scRNA–seq data. The accuracy of existing scRNA–seq clustering algorithms has not been comprehensively quantified, as the algorithms were applied to data sets in which the 'true' cell types were not independently known[7,9–11]. These methods have mainly been qualitatively validated by demonstrating that, on average, cells within a cluster tend to express the expected cell type markers. This marker-based approach can misrepresent cluster accuracy, as single-cell assays often fail to detect moderately expressed genes and may also detect ectopic expression of marker genes in unexpected cell types[18,19].

To benchmark existing clustering algorithms, we generated scRNA–seq data from 630 cells derived from seven cell lines, of which 561 passed quality control (**Fig. 1a**, Online Methods and **Supplementary Fig. 1**). To assess batch effects, we performed scRNA–seq in two batches for GM12878 (lymphoblastoid) cells and also for H1 embryonic stem cells. Gene expression was quantified as fragments per kilobase per million reads (FPKM), and low-quality cells were discarded on the basis of multiple metrics (**Fig. 1b** and Online Methods). The resulting expression table was supplied to eight existing scRNA–seq clustering algorithms (Online Methods): hierarchical clustering using all expressed genes (All-HC)[20], hierarchical clustering using principal-component analysis (PCA)-based feature selection (HiLoadG-HC)[7], BackSPIN[10], RaceID2 (ref. 9), Seurat[11] and three additional methods based on selection of genes with highly variable expression (VarG-HC, VarG-PCAproj-HC and VarG-tSNEproj-HC). We used the adjusted Rand index (ARI)[21] to quantify cluster accuracy: a

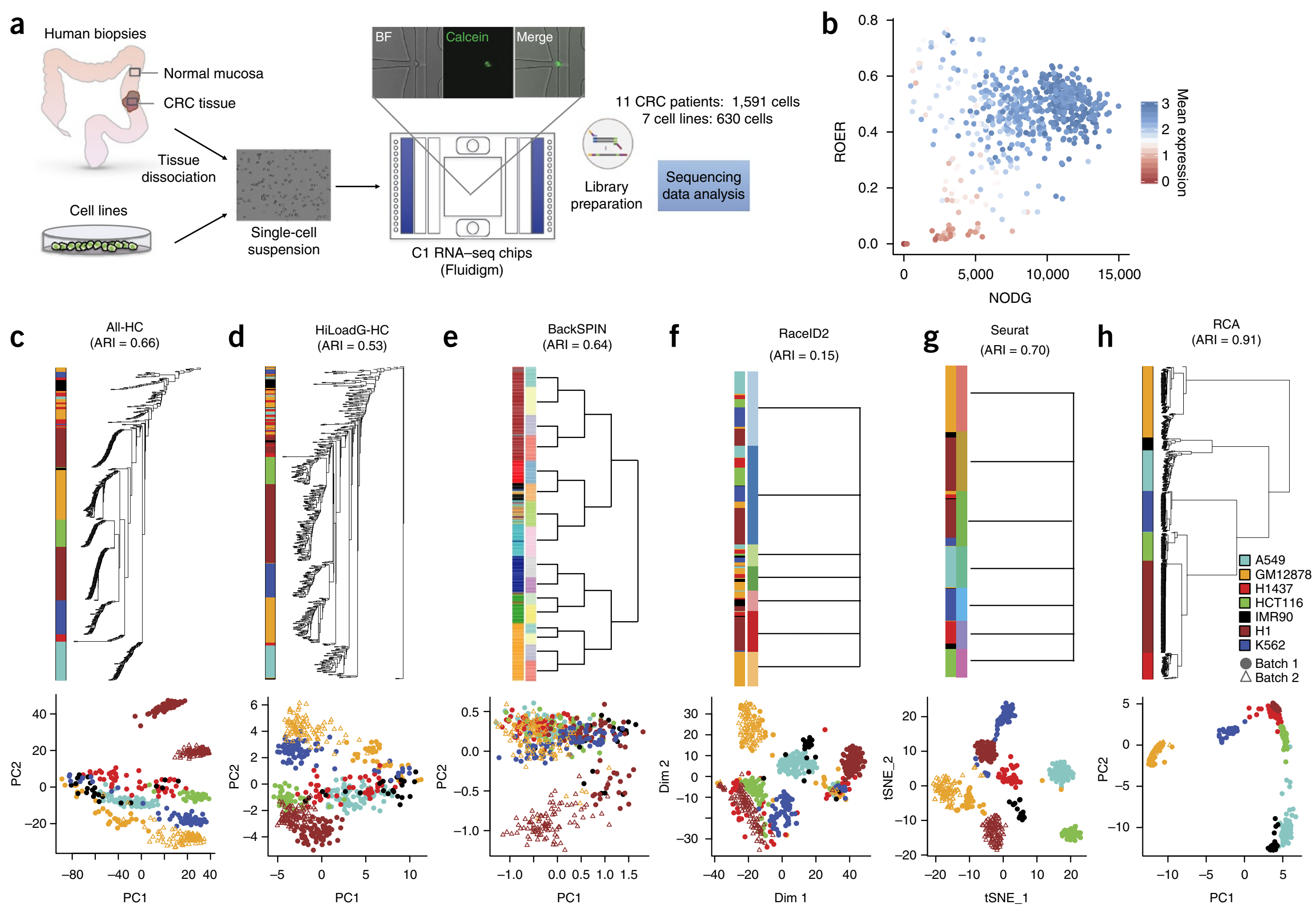


Figure 1 RCA outperforms existing algorithms for clustering single-cell transcriptomes. (**a**) Experimental workflow. BF, bright field. (**b**) Scatterplot of the number of detected genes (NODG) versus the rate of exonic reads (ROER) for the benchmarking cell line data. Colors correspond to the mean of housekeeping gene expression in $\log_{10}$ (FPKM). (**c**–**h**) Performance of different scRNA–seq clustering algorithms on the benchmarking data set. Colors correspond to the seven different cell lines used. Filled circles and open triangles represent two different batches of H1 and GM12878 data. The result of each algorithm was visualized by both dendrogram (top) and PCA plot (tSNE for RaceID2 and Seurat) (bottom). (**c**) All-HC method. (**d**) HiLoadG-HC method. (**e**) BackSPIN method. The dendrogram was generated manually on the basis of the biclustering results at different levels. (**f**) RaceID2 method. A one-level dendrogram was manually generated to show the results of *k*-medoids clustering. (**g**) Seurat. (**h**) RCA.

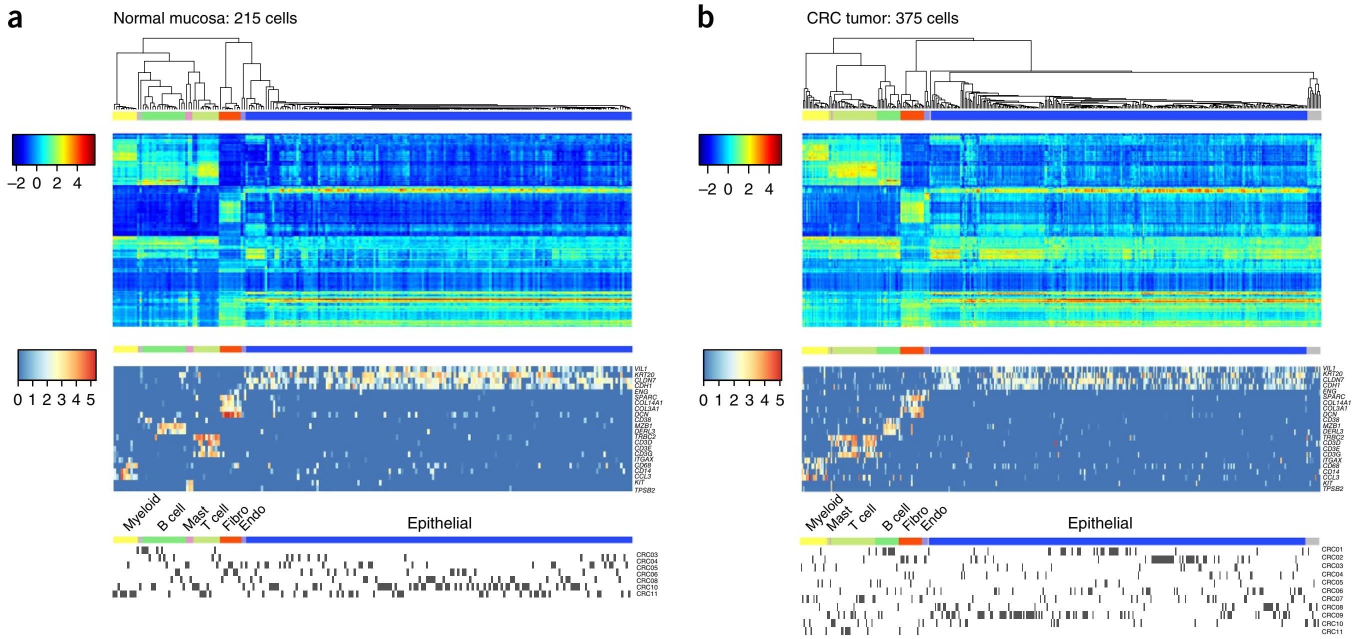


Figure 2 RCA identifies multiple cell types from CRC tumor and normal mucosa. (**a**) RCA clustering of cells from normal mucosa. (**b**) RCA clustering of cells from CRC tumors. In **a** and **b**, the dendrogram shows cell type clusters; in the upper heat map, columns represent cells, rows represent the reference panel and colors represent the projection scores on the reference panel; the middle heat map shows the raw $\log_{10}$ (FPKM) values for known markers of epithelial cells (*VIL1*, *KRT20*, *CLDN7*, *CDH1*), endothelial cells (Endo; *ENG*), fibroblasts (Fibro; *SPARC*, *COL14A*, *COL3A1*, *DCN*), B cells (*CD38*, *MZB1*, *DERL3*), T cells (*TRBC2*, *CD3D*, *CD3E*, *CD3G*), myeloid cells (*ITGAX*, *CD68*, *CD14*, *CCL3*) and mast cells (*KIT*, *TPSB2*); in the lower heat map, rows represent different patients and black-colored blocks in each row indicate cells from the corresponding patient.

value of 0 indicates random clustering, while a value of 1 represents perfect concordance.

Although All-HC segregated most cells accurately into distinct clusters, approximately 15% of cells were clustered incorrectly (ARI = 0.66; **Fig. 1c**). Moreover, both H1 and GM12878 cells were split into batch-specific clusters. HiLoadG-HC had lower performance (ARI = 0.53) with a substantial number of cells incorrectly clustered, and H1 and GM12878 cells again clustered by batch (**Fig. 1d**). BackSPIN, which is based on biclustering of variably expressed genes, had a similar ARI to All-HC (**Fig. 1e**). Batches of cells formed distinct clusters, although these clusters converged at a higher level in the clustering dendrogram. RaceID2 was far less accurate (ARI = 0.15 for the *k*-medoids step, 0.14 for the final step) and also more strongly influenced by batch effects (**Fig. 1f**). Seurat provided a marginal improvement over All-HC (ARI = 0.70) but was nevertheless vulnerable to batch effects and misclassified a substantial amount of cells (**Fig. 1g**). The three additional approaches based on variable gene selection yielded no improvement in accuracy (ARI = 0.0005–0.13; **Supplementary Fig. 2**). Given the limited accuracy of existing clustering algorithms, we hypothesized that technical artifacts (rather than biological differences) could explain much of the systematic variation in scRNA–seq data. Indeed, we found that a known technical covariate[12,13], the number of detected genes (NODG) in each cell, accounted for almost all of the variance in the first principal component of the expression matrix (**Supplementary Fig. 3a**).

The above results indicate that new clustering approaches are needed for accurate cell type identification from scRNA–seq data. One possibility would be to emulate the reference projection strategy recently developed for ancestry estimation from low-coverage whole-genome sequencing[22]. This approach projects noisy, incomplete data sets onto a lower-dimensional space defined by variability in an independent, higher-quality reference panel. To construct an analogous reference panel for transcriptomes, we compiled bulk-sample transcriptomes spanning diverse tissue and cell types (global panel; Online Methods).

scRNA–seq data sets were projected onto the global reference panel, and normalized coordinates in the projection space were supplied to a conventional hierarchical clustering program to define cell type clusters (Online Methods). We tested this method, which we named RCA, on the above benchmark. Notably, RCA generated tight cell clusters consisting almost entirely of cells of the same type (**Fig. 1h**). Moreover, cells of the same type clustered together even when profiled in different batches. The ARI (0.91) was far greater for RCA than for existing clustering algorithms. We also evaluated RCA and the other eight algorithms on an independent data set from scRNA–seq profiling of melanoma[6]. Again, RCA provided nearly perfect accuracy while the other eight algorithms had lower ARI values (Online Methods, **Supplementary Fig. 4** and **Supplementary Table 1**). These results demonstrate that RCA greatly improves the accuracy of single-cell transcriptome clustering.

RCA identifies multiple cell types in CRC tumors and normal mucosa

We used scRNA–seq to profile 969 cells from the resected primary tumors of 11 patients with CRC. For comparison, we also profiled 622 single cells from the nearby normal mucosa of seven of the patients (**Fig. 1a**, Online Methods and **Supplementary Table 2**). After stringent filtering for data quality (Online Methods), 375 tumor cells and 215 normal mucosa cells were retained. We first examined the cell type composition of the colonic normal mucosa. RCA analysis using the global panel yielded seven distinct cell clusters (**Fig. 2a** and **Supplementary Fig. 5a**). Note that each normal mucosa cell cluster spanned multiple data batches (multiple patients)—we did not observe clustering by patient. On the basis of the expression of known markers, the seven clusters were annotated as epithelial cells, fibroblasts, endothelial cells, B cells, T cells, mast cells and myeloid cells (**Fig. 2a** and

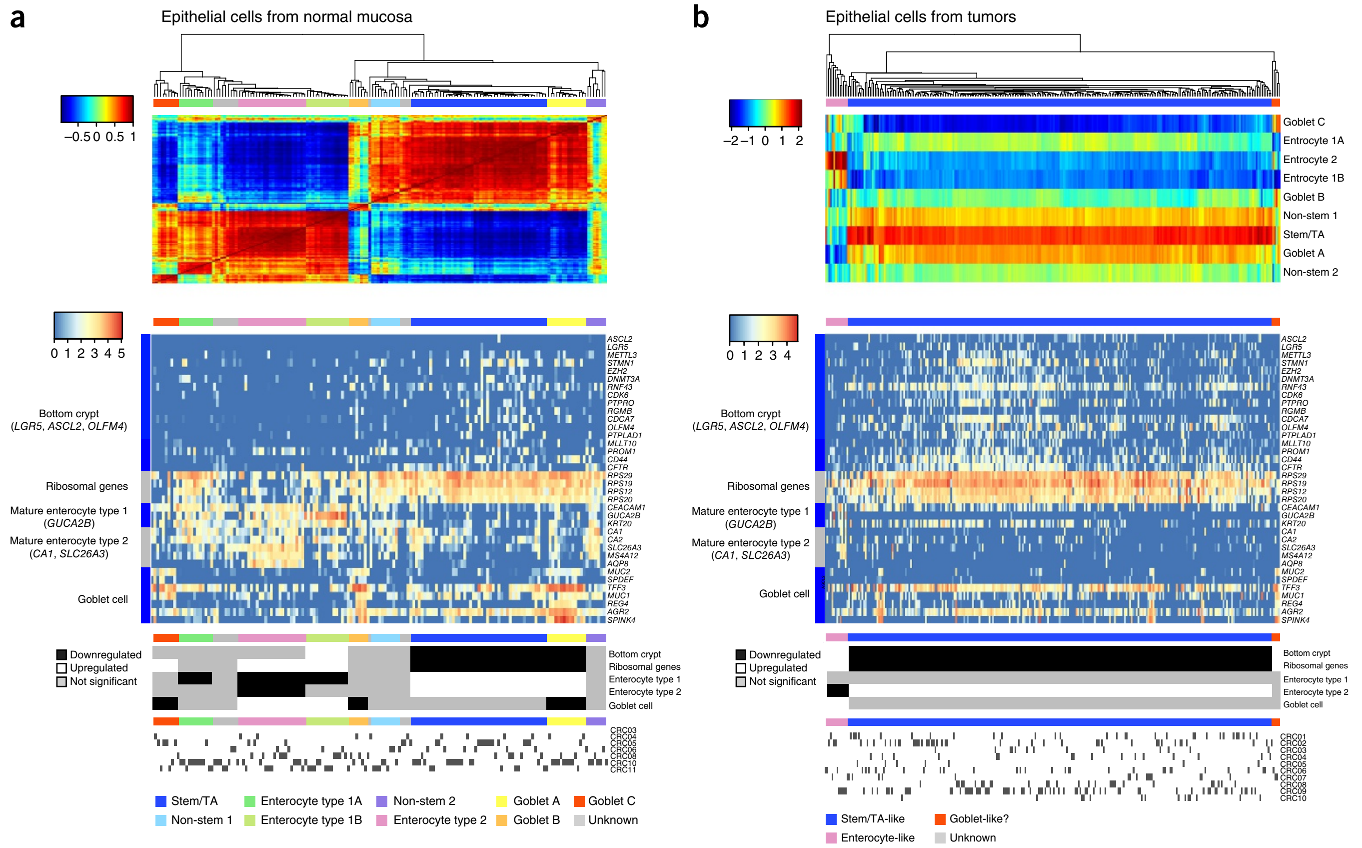


Figure 3 RCA identifies epithelial cell subtypes from CRC tumors and normal mucosa. (**a**) RCA clustering of cells from normal mucosa using self-projection (Online Methods). (**b**) RCA clustering of cells from CRC tumors using projection onto the colon epithelium panel (Online Methods). In **a** and **b**, the dendrogram shows cell type clusters, the heat map in the center shows the $\log_{10}$ (FPKM) values of cell type markers, the lower heat map shows up- or downregulation of cell type metagenes in individual clusters, and the black boxes at the bottom indicate the patient ID for each cell. In **a**, the uppermost heat map shows the self-projection scores of normal mucosa epithelial cells (Online Methods). In **b**, the uppermost heat map shows the projection scores of tumor epithelial cells onto the colon epithelium panel (Online Methods).

Online Methods). These seven cell types were also identified by RCA clustering of CRC tumor cells (**Fig. 2b** and **Supplementary Fig. 5b**). Thus, our strategy of profiling unsorted cells followed by RCA clustering facilitated the capture and identification of a broad range of cell types from CRC and normal mucosa samples.

In the above analysis, normal epithelial cells did not form distinct subclusters, indicating that epithelial subtypes such as enterocytes, goblet cells and transit-amplifying (TA) cells were not well resolved (**Fig. 2**). We therefore tested a variant of RCA clustering in which the epithelial single-cell transcriptomes were used as their own reference panel (self-projection; Online Methods). This approach yielded nine subclusters of normal epithelial cells. As above, we did not observe subclustering by batch (**Fig. 3a**). The subclusters were also supported by the expression patterns of epithelial markers, suggesting that they represented distinct cell types and/or states (**Fig. 3a** and Online Methods).

One of the epithelial subclusters in normal mucosa was annotated as stem/TA on the basis of elevated expression of bottom-crypt markers[23–25] and ribosomal genes[8]. Three subclusters (goblet A, goblet B and goblet C) expressed goblet cell markers such as *TFF3*, *MUC2* and *SPDEF*. Goblet A cells showed a TA signature, as well as strong expression of goblet cell markers. Goblet C cells expressed goblet cell markers but no TA signature, suggesting a more mature phenotype. We identified two enterocyte subtypes, consistent with previous findings[23]: GUCA2B$^+$ (enterocyte type 1; **Fig. 3a**) and SLC26A3$^+$ (enterocyte type 2). These were further subdivided into clusters of cells reflecting variation in maturation state. The epithelial clusters identified by RCA self-projection were validated by analysis of differentially expressed genes (**Supplementary Fig. 6** and **Supplementary Table 3**) and RNAscope analysis (**Supplementary Fig. 7**).

To anchor the analysis of tumor epithelial cell states, we constructed a novel reference panel of nine epithelial transcriptomes from normal mucosa (colon epithelium panel; Online Methods). On the basis of this reference panel, RCA identified three clusters of tumor epithelial cells: stem/TA-like, enterocyte 2B–like and goblet-like (**Fig. 3b**). Stem/TA-like cells accounted for 93% of tumor epithelial cells, in contrast to 30% of normal mucosa epithelium, consistent with the proliferative nature of CRC cells. Thus, RCA robustly identified cell types in normal mucosa and CRC tumors, despite the strong batch effects in clinical samples.

scRNA–seq data analysis identifies genes differentially expressed in tumors

We hypothesized that tumor–normal differentially expressed genes identified from single-cell data would differ substantially from differentially expressed genes detected using more traditional bulk-sample comparisons. For single-cell analysis of differential expression, we prioritized the five largest cell clusters from tumor samples: stem/TA-like epithelial cells, fibroblasts, B cells, T cells and myeloid cells (Online Methods). Each of these tumor cell clusters was compared to the corresponding normal mucosa cluster (**Figs. 2b** and **3b**), resulting in 129 differentially expressed genes at false discovery rate (FDR) ≤ 0.05 (**Supplementary Fig. 8** and **Supplementary Table 4**), largely contributed by stem/TA-like cells and fibroblasts. We compared these single-cell differential expression results to those from bulk-sample tumor–normal analysis of a large CRC cohort[26] and found that the fold-change direction was significantly concordant for stem/TA-like cells ($P = 2.97 \times 10^{-5}$) and fibroblasts ($P = 8.77 \times 10^{-4}$) (Fisher's exact test; **Fig. 4a,b**). Moreover, single-cell lists of differentially expressed genes included numerous genes also highlighted by previous functional studies of CRC (**Supplementary Note**). Nevertheless, the majority of differentially expressed genes from single-cell analysis were undetected in bulk analysis (**Fig. 4a–d**). In summary, the concordance between differentially expressed genes identified by single-cell and bulk analyses was significant but limited.

One drawback of bulk-sample differential expression analysis is that it could classify a cell-type-specific gene as dysregulated in cancer merely because of a systematic difference in cellular composition. For example, tumors may have a high number of stem/TA-like cells, thus causing most markers of this cell type to appear upregulated in bulk-tumor analysis (**Supplementary Fig. 9** and **Supplementary Table 5**). To investigate such biases, we used normal mucosa single-cell data to identify 292 genes differentially expressed among the major cell types or among the epithelial subtypes (**Fig. 4e** and Online Methods). After removing the 48 genes that were also differentially expressed between tumor and matched normal single cells, we were left with 244 marker genes. These 244 genes were significantly enriched in the bulk-sample differential expression set relative to the entire set of expressed genes ($P = 2.1 \times 10^{-10}$, Fisher's exact test). In fact, all four of the genes differentially expressed in the bulk sample at $P \leq 1 \times 10^{-22}$ were cell type markers ($P = 5.7 \times 10^{-3}$, Fisher's exact test; **Fig. 4f**). These results are consistent with contamination of the list of differentially expressed genes from bulk-sample analysis by cell type markers.

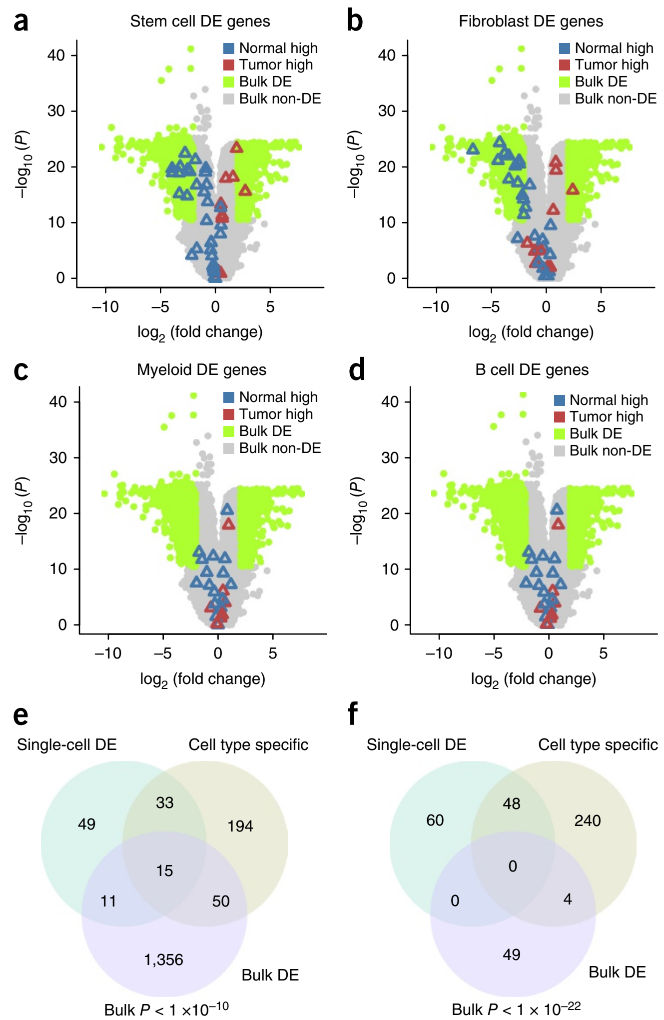


Figure 4 scRNA–seq data analysis identifies novel differentially expressed genes between tumor and normal tissues. (**a**–**d**) Differentially expressed (DE) genes between normal stem/TA cells and tumor stem/TA-like cells (**a**), between normal fibroblasts and tumor fibroblasts (**b**), between normal and tumor myeloid cells (**c**), and between normal and tumor B cells (**d**). Differentially expressed genes from single-cell data are highlighted in volcano plots of bulk differential expression analysis. Green color represents bulk differentially expressed genes; gray represents bulk non–differentially expressed genes. Red triangles represent upregulated single-cell differentially expressed genes; blue triangles represent downregulated single-cell differentially expressed genes. (**e**,**f**) Venn diagrams showing the overlap between differentially expressed genes from bulk and single-cell analyses and cell-type-specific genes. The *P*-value cutoffs for differentially expressed genes in bulk analysis were 1×10^{-10} for **e** and 1×10^{-22} for **f**.

scRNA–seq data analysis highlights pathway alterations and diversity of CAFs in CRC

To identify upstream regulators of the differentially expressed genes in single-cell analysis, we used Ingenuity Pathway Analysis (IPA)[27]. Results from four cell types (stem/TA-like cells, fibroblasts, myeloid cells and B cells) were merged and ranked by *P* value. The top three upstream regulators included two small molecules: actinonin and motexafin gadolinium (**Fig. 5a**). Actinonin inhibits growth in multiple cancer cell lines, including colon cancer[28,29]. In our analysis, actinonin was predicted to antagonize the expression changes detected in tumor-associated myeloid cells, suggesting that it may also act on tumor immune cells[30]. Motexafin gadolinium, which was the top hit in stem/TA-like epithelial cells, is known to induce apoptosis in cancer cells[31].

TGFB1 was the top ranked endogenous upstream regulator, with a positive *z* score indicating increased pathway activation in CAFs (**Fig. 5a**). Consistent with this, the transforming growth factor (TGF)-β effector SMAD3 was also predicted to have increased activity in CAFs. In addition, *TGFB3* and the *LTPB1* gene encoding a TGF-β-binding protein were more frequently expressed in CAFs than in normal mucosa fibroblasts (**Fig. 5b,d**). Immunohistochemical (IHC) analysis confirmed the increased protein level of TGF-β in the CAF compartment (**Fig. 5c**). In normal mucosa, TGF-β-responsive genes showed expression mainly in myeloid cells. In contrast, in tumors, TGF-β-responsive genes such as *TGFBI*, *CTGF* and *BHLHE40* were mainly expressed in epithelial cells and CAFs (**Fig. 5d** and **Supplementary Fig. 10**). These results suggest that TGF-β secretion by CAFs may contribute to TGF-β pathway activation in tumor epithelial cells.

Intriguingly, CAFs showed bimodal expression of some TGF-β pathway genes (**Fig. 5d**). We therefore used RCA in self-projection mode to cluster CAFs and normal mucosa fibroblasts and identified three clusters of fibroblast cells (**Fig. 5e,f** and **Supplementary Fig. 11a**). One cluster comprised mainly normal fibroblasts (normal mucosa fibroblasts, NMFs), while the other two (CAF-A and CAF-B) contained mostly CAFs. Differential expression analysis showed that CAF-B cells expressed markers of myofibroblasts such as *ACTA2*, *TAGLN* and *PDGFA* (**Fig. 5f** and **Supplementary Fig. 11b**). These markers were downregulated in CAF-A cells, which expressed *MMP2*, *DCN* and *COL1A2*. The two CAF subtypes were detected in scRNA–seq data from multiple patients (**Fig. 5f**). In addition, we confirmed

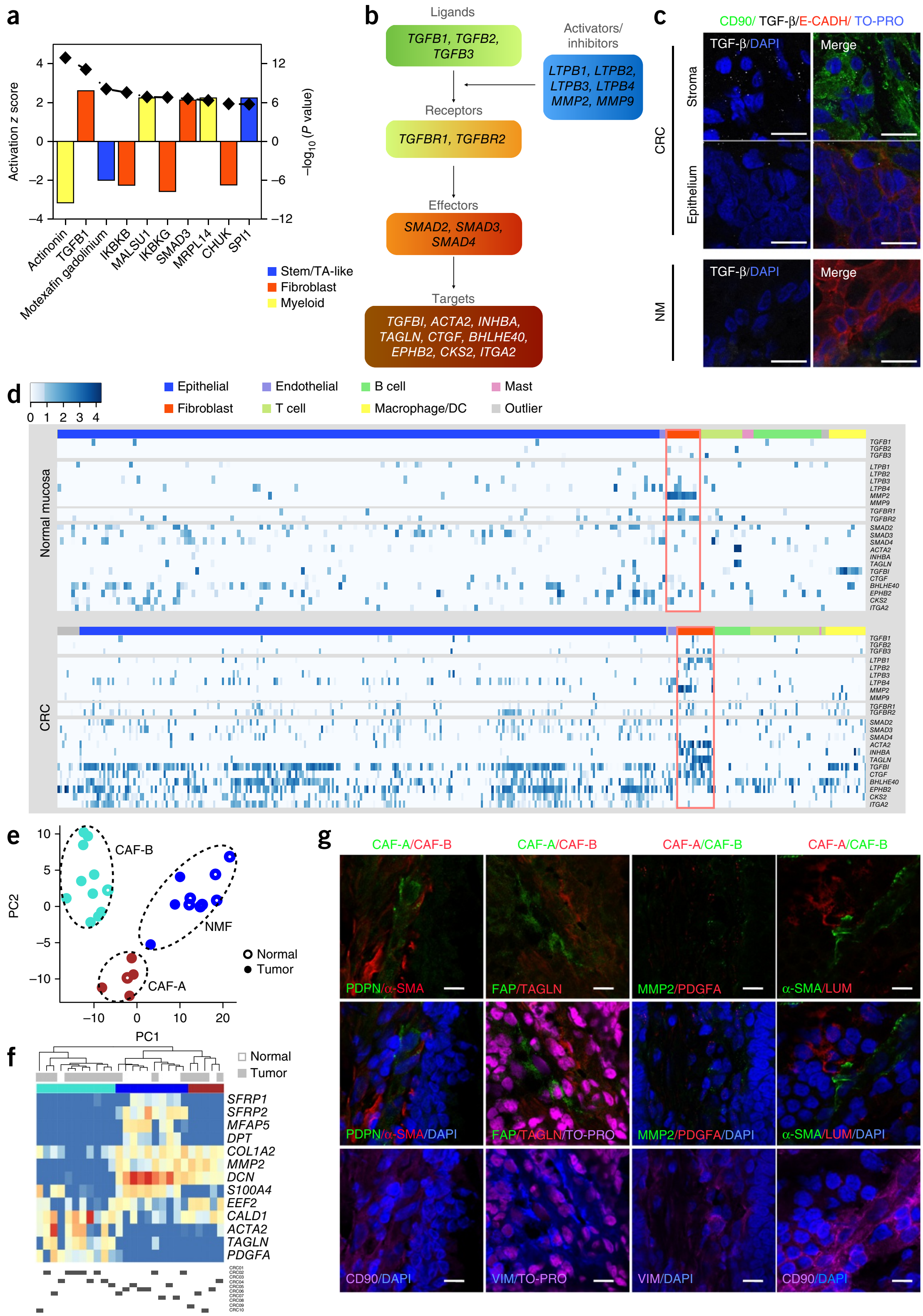


Figure 5 scRNA–seq data highlight pathway alterations and diversity of CAFs in CRC. (**a**) Activation *z* scores and *P* values of the top predicted upstream regulators from IPA analysis of stem/TA-like cells, fibroblasts and myeloid cells. (**b**) Components of the TGF-β signaling pathway. (**c**) Detection of TGF-β (red) in epithelial cells (E-cadherin+; blue) and fibroblasts (CD90+; green) by IHC in normal mucosa (NM) and CRC tissue. Nuclei were stained with TO-PRO. Scale bars, 10 μm. (**d**) Raw $\log_{10}$ (FPKM) values of genes involved in the TGF-β signaling pathway in major cell types of the normal mucosa and CRC tissues. DC, dendritic cell. (**e**) PCA plot of CAF subtypes. (**f**) CAF subtypes presented in a heat map of differentially expressed genes. The black boxes at the bottom indicate the patient ID for each cell. (**g**) Validation of CAF subtypes by immunofluorescence in CRC tissue sections. CAFs were counterstained for CD90 or vimentin (VIM). Coexistence of the CAF-A and CAF-B subtypes was confirmed in four primary tumors. Scale bars, 10 μm.

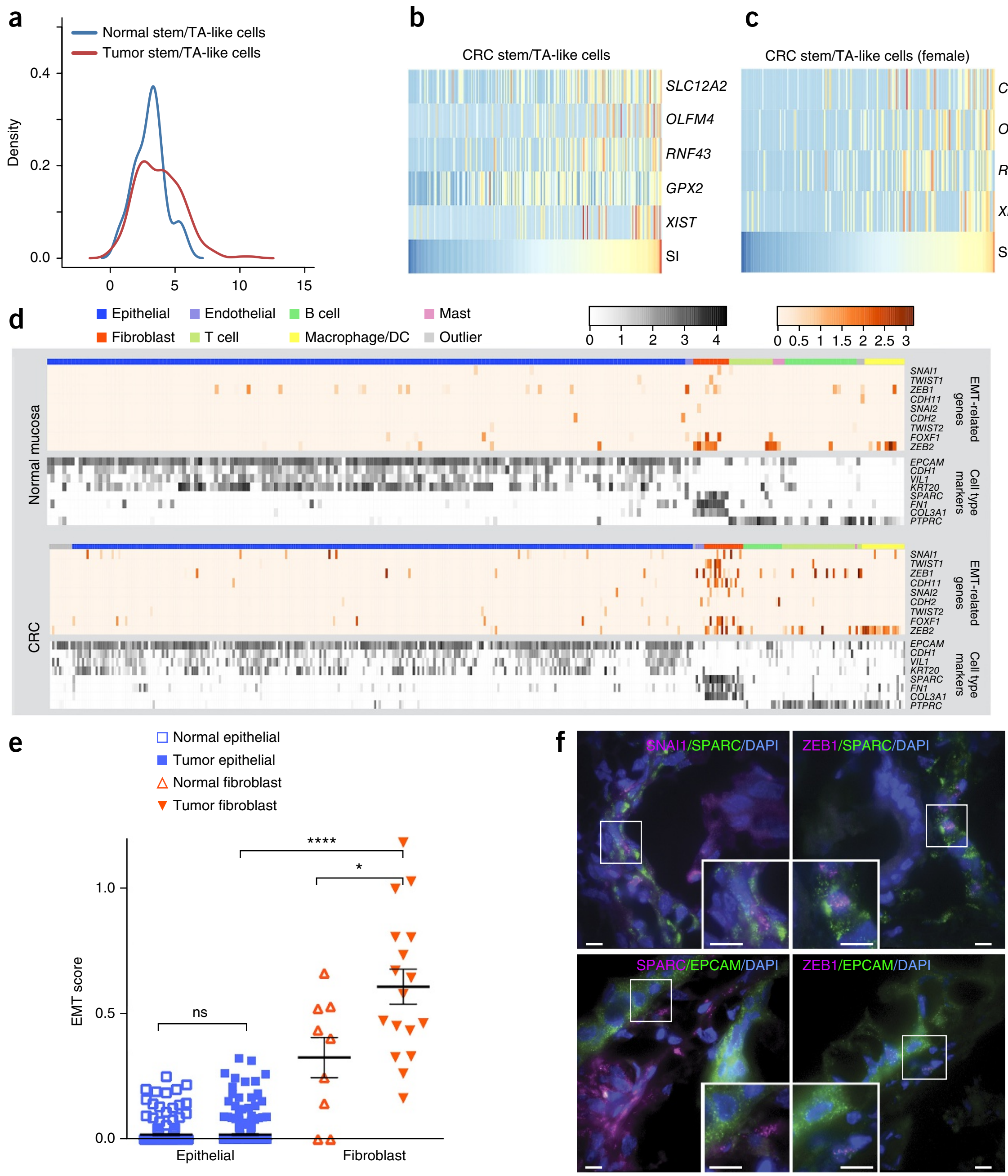
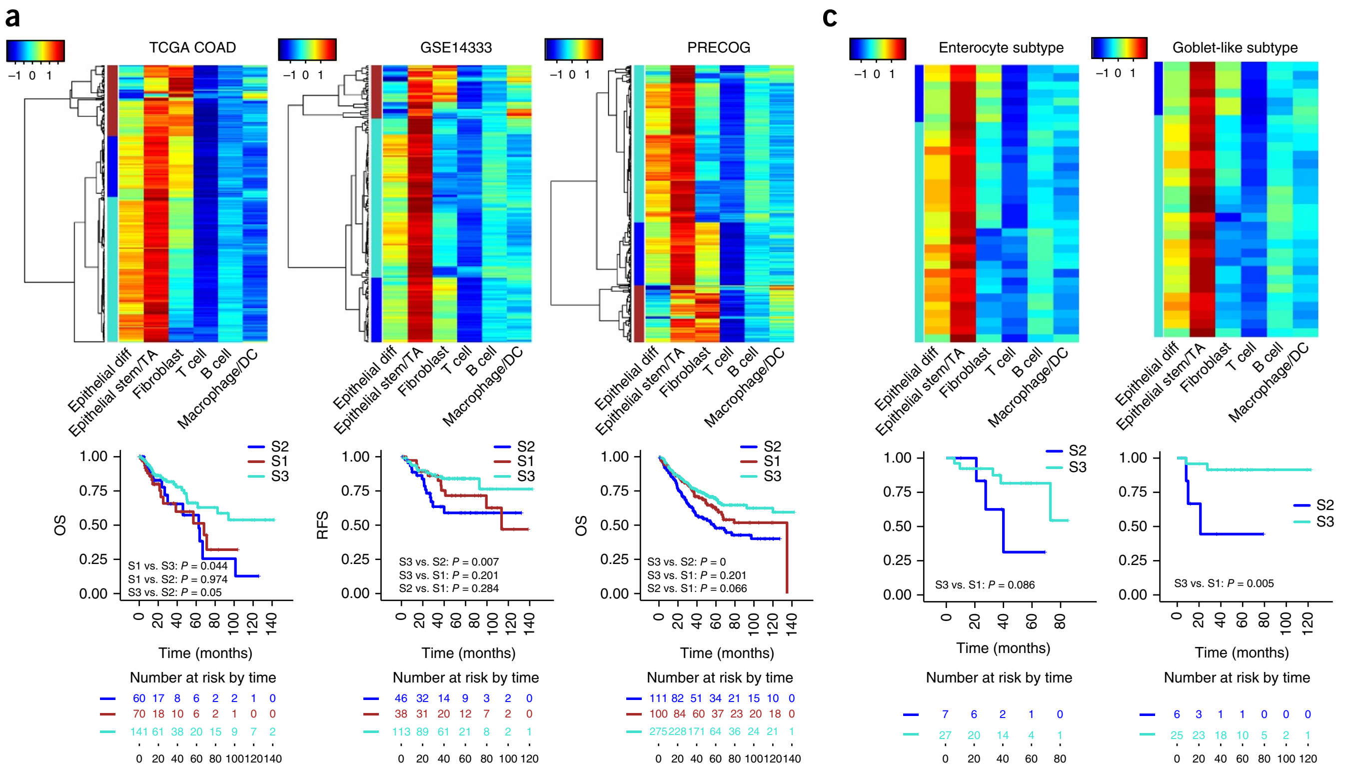


Figure 6 Molecular phenotyping of single cells: stemness and EMT. (**a**) Distribution of stemness index (SI) values for stem/TA-like cells in normal mucosa and CRC tumors. (**b**,**c**) *z* scores of $\log_{10}$ (FPKM) values for genes whose expression is highly correlated with SI across all tumor stem/TA-like cells (**b**) and those from female patients only (**c**). (**d**) Raw $\log_{10}$ (FPKM) values of EMT-related transcription factors (orange). Expression of cell type markers is depicted in grayscale below. (**e**) EMT scores of epithelial and fibroblast cells in normal mucosa and CRC. Center values represent the mean, and error bars denote s.e.m. $*P < 0.05$, $****P < 0.0001$ (Wilcoxon rank-sum test); ns, not significant. (**f**) smFISH validation of EMT markers in tumor sections. Scale bars, 10 μm.

their coexistence in tumor sections from four patients using seven subtype-specific markers (**Fig. 5g**). Thus, CAFs within CRC tumors comprise two major subtypes with distinct transcriptomic profiles.

Molecular phenotyping of single cells: stemness and EMT

Tumor stem/TA-like cells appeared to express higher levels of bottom-crypt markers than corresponding cells from normal mucosa (**Fig. 3**). To quantify this observation, we defined a stemness index (SI) by averaging the expression levels of 42 known markers of colon or intestinal stem cells (**Supplementary Table 6**). The distribution of SI values was continuous, both in tumors and normal mucosa (**Fig. 6a**). Tumor cells had significantly higher SI scores ($P = 0.01$, Wilcoxon rank-sum test), and 5% of tumor stem/TA-like cells had scores beyond those found in normal mucosa cells. By correlating gene expression with SI score across stem/TA-like cells, we identified five stemness-associated genes (Pearson correlation ≥ 0.35; **Fig. 6b**). Consistent with the known role of WNT signaling in colorectal cancer[32], these genes included three WNT mediators: *GPX2*, *OLFM4* and *RNF43* (**Supplementary Fig. 12**). Intriguingly, the long noncoding RNA (lncRNA)-encoding gene *XIST*, which mediates female X-chromosome inactivation, was also identified



b

Gene signature	S1	S2	S3
Epithelial differentiated	–	+	++
Epithelial stem/TA	+	++	+++
Fibroblast	+++	+	–
T cell	–	–	–
B cell	–	–	–
Macrophages/DC	++	–/+	–

Figure 7 Single-cell transcriptome signatures stratify CRC tumors into subgroups with distinct patient survival. (**a**) The 'fibroblast-like' signature is correlated with poor patient survival. In the clustering heat maps, rows represent bulk tumor samples from distinct patients with CRC and columns represent the projection of tumor samples onto the six most abundant cell type populations from scRNA–seq data. The colors for the survival curves match those used for the S1, S2 and S3 subgroups in the heat maps. OS, overall survival; RFS, relapse-free survival. (**b**) Table summarizing the contribution of each cell type signature in the S1, S2 and S3 subgroups. (**c**) Bulk projection heat maps and survival analysis of the enterocyte and goblet-like tumors from the GSE14333 cohort according to the previously published CRCA system[34].

as associated with stemness. This finding was recapitulated when we analyzed female tumors exclusively (**Fig. 6c**). In contrast, XIST expression was not correlated with SI score in stem/TA-like cells from normal mucosa samples (**Supplementary Fig. 13a,b**). In summary, the stemness signature shows continuous variation across stem/TA-like tumor cells and correlates with the expression of XIST and WNT mediators.

It has been proposed that epithelial cells transition to a mesenchymal state in tumors, which permits detachment and migration to new sites and thus facilitates metastasis. Transcriptionally, EMT is defined by downregulation of the epithelial marker *CDH1* and upregulation of mesenchymal transcription factors from the TWIST, SNAIL and ZEB families. To investigate the prevalence of EMT in CRC, we examined the expression of these genes, alongside that of other cell type markers (**Fig. 6d**). Notably, we did not observe differential expression of *CDH1* between tumor and normal mucosa epithelium. Moreover, neither tumor nor normal mucosa epithelial cells expressed EMT transcription factors at appreciable levels. Rather, some EMT transcription factors appeared to be upregulated in CAFs. This expression change is unrelated to EMT, as fibroblasts are mesenchymal cells. To quantify these observations, we defined a per-cell EMT score by combining the expression values of the nine EMT-upregulated genes in **Figure 6d**. This analysis confirmed the lack of an EMT signature in tumor epithelial cells, as well as the significantly increased EMT signature in CAFs (**Fig. 6e**).

To rule out tissue dissociation artifacts, we visualized mRNA molecules in tumor sections using single-molecule FISH (smFISH). Expression of the EMT markers *SNAI1*, *ZEB1* and *TWIST1* coincided with that of the fibroblast marker *SPARC* and could not be detected in *EPCAM*-expressing epithelial cells (**Fig. 6f** and **Supplementary Fig. 14b**). These findings were further supported by tissue immunofluorescence and bioinformatics analysis of known epithelial and fibroblast markers (**Supplementary Figs. 14a** and **15**, and **Supplementary Note**).

Single-cell transcriptome signatures stratify CRC tumors into subgroups with distinct patient survival

Colorectal tumors have been classified on the basis of bulk transcriptomes into subtypes that differ in treatment response and patient survival[33–36]. These classification systems are influenced by variability in the transcriptomes of stromal cell types[16,17]. We therefore explored the contributions of the major cell types within CRC tumors to bulk transcriptomic signatures and patient survival. We constructed the expression signatures of six tumor cell types by averaging across the corresponding single-cell transcriptomes (Online Methods). We then compiled bulk-transcriptomic data from six independent cohorts annotated with survival information: TCGA, GSE14333 (ref. 34), PRECOG37 (ref. 37) and three cohorts from a recent consortium study[38]. We projected the bulk transcriptomes onto the six cell type signatures and clustered them in this space (**Fig. 7a**, Online Methods and **Supplementary Fig. 16**). We identified the same three tumor groups in each of the six cohorts: S1, S2 and S3 (**Fig. 7a**). S1 tumors had a weak epithelial signature, a strong fibroblast signature and an elevated myeloid signature. S2 tumors had intermediate levels of all signatures, while S3 tumors had a strong epithelial signature and weak fibroblast and myeloid signatures (**Fig. 7b**).

We then explored the prognostic significance of the newly defined CRC subtypes. In all six cohorts, S3 tumors had the best survival rate (**Fig. 7a** and **Supplementary Fig. 16**). This is consistent with previous studies associating fibroblast signatures with patient outcome[16,17,39,40]. We noticed that the 'enterocyte' and 'goblet-like' CRC tumor subtypes previously defined[34] on the basis of bulk transcriptomes from GSE14333 included both S2 and S3 subtypes from our analysis (**Fig. 7c**). Within the goblet-like subtype, S3 tumors had significantly higher survival rates than S2 tumors ($P = 0.005$). The same disparity was observed within the enterocyte subtype, although in this case the difference was not statistically significant. These results further underscore the potential relevance of CAFs to CRC outcome and demonstrate the prognostic value of the cell-type-specific signatures inferred from single-cell transcriptomes.

DISCUSSION

We have performed an unbiased single-cell transcriptomic analysis of intratumoral heterogeneity in CRC, covering both cancer and stromal cells. In addition, we generated a comprehensive benchmark data set of 630 single-cell transcriptomes from seven cell lines to evaluate the accuracy of scRNA–seq clustering algorithms. Unexpectedly, existing algorithms were unable to accurately cluster the 630 cells by cell type (**Fig. 1c–g** and **Supplementary Figs. 2** and **4**), even though the cell line data set was of higher quality than would be expected from primary tissue samples (**Fig. 1b** and **Supplementary Fig. 3c**). The limited accuracy of these methods could potentially reflect their propensity to interpret batch effects and technical variation as biological variation in single-cell transcriptomic data[7,8,10,12,13] (**Supplementary Fig. 3a**,**b**). We therefore developed a novel algorithm named reference component analysis (RCA) that projects single-cell transcriptomes into a space defined by variability in a reference data set. RCA clustered the benchmark data with high accuracy, indicating that reference-guided clustering can greatly reduce technical variation and batch effects. This approach thus seems to be the most robust for identifying the major cell types in a sample. However, projection onto an external reference may not always detect subtle differences between cellular subtypes, such as epithelial cells in diverse differentiation states. At present, the self-projection variant of RCA would therefore be more appropriate for detecting subtypes or distinct cell states within a major cell type. However, as available reference data sets expand in size and diversity, we anticipate that the resolution of reference-guided RCA will continuously improve.

We identified seven major cell types using RCA, both in normal mucosa and CRC tumors (epithelial cells, fibroblasts, endothelial cells, B cells, T cells, mast cells and myeloid cells). This marker-free method for purifying cell types *in silico* permitted unbiased analysis of cell-type-specific genes differentially expressed between tumor and normal samples. Furthermore, we used RCA to identify nine epithelial clusters and to define seven epithelial cell subtypes *de novo* in human normal mucosa. We used the nine normal mucosa epithelial subclusters as a reference to infer the differentiation state of tumor epithelial cells, showing strong enrichment of stem/TA-like cells in tumor samples (**Fig. 3**). In addition to elucidating CRC functional states, we anticipate that the normal mucosa scRNA–seq data set could also serve as a valuable resource in future studies of human colorectal biology and related pathologies.

Traditionally, dysregulated genes in tumors have been identified using bulk transcriptomics. However, this approach could be confounded by differences in cell type proportion between diseased and normal tissues[41]. Indeed, we observed profound differences between our lists of cell-type-specific differentially expressed genes and the differentially expressed genes previously detected from bulk tumor–normal analysis of CRC (**Fig. 4a–d**), and the differentially expressed genes from bulk analysis were more likely to be differentially expressed between diverse cell types within normal mucosa than between tumor and normal mucosa cells of the same type (**Fig. 4e**,**f**). We therefore propose that single-cell transcriptomics followed by cell type clustering provides a more unbiased approach for identifying dysregulated genes within tumor and stromal cells.

Another advantage of the unbiased single-cell approach is the ability to investigate cell-type-specific expression of genes related to pathways of interest to tumor biology. Pathway enrichment analysis and gene panel analysis both highlighted enhanced TGF-β signaling in CRC[42,43], which is potentially driven by crosstalk between epithelial and stromal cells (**Fig. 5**). Single-cell transcriptomics and IHC analysis suggest that CAFs may contribute to TGF-β signaling by secreting TGF-β3 ligand and the TGF-β activator MMP2. TGF-β-responsive genes were upregulated in CAFs as well as tumor epithelial cells, suggesting that both autocrine and paracrine responses may be present within the tumor.

RCA clustering elucidated the heterogeneity of the CAF population. We identified two distinct subpopulations of tumor fibroblasts, named CAF-A and CAF-B. CAF-B cells expressed cytoskeletal genes and other known markers of activated myofibroblasts, while CAF-A cells expressed genes related to extracellular matrix remodeling, including the TGF-β activator *MMP2*. CAF-A could represent an intermediate state between normal fibroblasts and the CAF-B/myofibroblast state or, alternatively, an independent CAF subtype. Although CAF heterogeneity has been noted in multiple studies[44–46], this is, to our knowledge, the first *de novo* identification of CAF subtypes on the basis of scRNA–seq data analysis. Multiple clinical trials have been based on targeting cells expressing FAP (fibroblast activation protein α); a membrane serine protease expressed only in CAFs) both directly and indirectly[47,48], but efficacy has not been demonstrated. Our data show that only CAF-A cells expresses FAP. Thus, CAF heterogeneity could potentially constitute an impediment to FAP-directed therapies.

Although EMT has been studied extensively *in vitro* and in model organisms[49], evidence for this phenomenon in human tumors has been limited[50,51]. In fact, two recent studies have shown that the upregulation of EMT-related genes in bulk colorectal tumors is driven by expression changes in fibroblasts, rather than in epithelial cells[16,17]. Consistent with these studies, we found no evidence for upregulation of EMT-related genes in tumor epithelial cells, although we detected significantly increased expression of EMT markers in CAFs. This could be a consequence of the transition to an activated fibroblast state, which is known to coincide with expression of EMT markers in CAFs[52]. Taken together, these results suggest that the prevalence of EMT may be low in colorectal tumors. However, it is nevertheless conceivable that EMT may occur under specific conditions or in specific subregions of colorectal tumors that were not sampled in our study.

CRC tumor classification based on bulk-transcriptome data is well established and has been shown to predict patient survival[33–36]. We found that an existing CRC classification system could be refined by projecting bulk-tumor transcriptomes onto cell-type-specific transcriptomes. In particular, tumors previously assigned to the enterocyte and goblet-like classes[34] could be further subdivided into groups (S2 and S3) with different survival outcomes (**Fig. 7c**). Thus, in addition to serving as a robust strategy for clustering single-cell transcriptomes, the reference projection strategy (RCA) can be used to stratify patients by their bulk-tumor transcriptomes. As the cost of single-cell transcriptomic assays continues to drop, one can envisage future CRC classification systems involving the generation of multiple transcriptomes from a single tumor, at the level of either single cells or sorted cell types.

URLs. Gene Expression Omnibus, https://www.ncbi.nlm.nih.gov/geo/; European Genome-phenome Archive, https://www.ebi.ac.uk/ega/; BackSPIN v1.0, https://github.com/linnarsson-lab/BackSPIN/releases; command list for Seurat (provided by its authors), https://www.dropbox.com/s/n3t13kk4mmo1ggs/pancreas.R?dl=1; Human U133A/GNF1H Gene Atlas and Primary Cell Atlas from BioGPS, http://biogps.org/; bulk RNA–seq data for CRC from the Cancer Genome Browser, https://tcga.xenahubs.net/download/TCGA.COAD.sampleMap/HiSeqV2; RCA R package, https://github.com/GIS-SP-Group/RCA.

METHODS

Methods, including statements of data availability and any associated accession codes and references, are available in the online version of the paper.

Note: Any Supplementary Information and Source Data files are available in the online version of the paper.

ACKNOWLEDGMENTS

We would like to thank L. Suteja, C. Kang, S. Sudhagar, J. Sheik and M.N. Ramalingam for technical assistance, A. Brichkina (Institute of Molecular and Cell Biology, A*STAR) for providing the antibody to SMA, V. Sivakamasundari for guidance on single-cell protocols, M.H. Chew, R. Ten, W.J. Lim, J.H. Lai, C.Y. Ng and D. Koh for assistance with clinical sample collection, and Y. Hu, S. Ghosh, H. Kitano and D. Tan for feedback and scientific discussions. This study was supported by core funds from the Agency for Science, Technology and Research (A*STAR) and also by grant JCO1331CFG080 from A*STAR's Joint Council Office. P.R. acknowledges support from Agency of Science, Technology and Research grants IAF111091 and IAF111128 and associated in-kind contributions from industry partners Fluidigm Singapore and Becton Dickinson Holdings, respectively. S.L.K. and A.M.H. acknowledge the Strategic Positioning Fund (SPF2012/003) from the Biomedical Research Council (BMRC).

AUTHOR CONTRIBUTIONS

H.L., E.T.C., L.J.K.W. I.B.T., P.R. and S.P. conceived the idea and designed the study. H.L. developed the computational algorithms and performed the bioinformatic analysis. E.T.C. optimized and conducted the experiments. D.S. assisted with the data analysis. Y.T. assisted with the experiments. J.J.L.G. and K.H.C. performed the smFISH experiments. S.L.K. assisted with the initial protocol validation. W.S.T. and L.K.H. extracted and preprocessed clinical samples and guided patient selection. C.C. coordinated the clinical sample registration, preprocessing and logistics. P.J.C. supported the smFISH experiments. A.H. provided guidance in experimental protocol design. H.L. and E.T.C. analyzed the data and interpreted the results. S.P. guided the development of computational algorithms. I.B.T., P.R. and S.P. provided guidance in data analysis and interpretation of the results. H.L., E.T.C., I.B.T., P.R. and S.P. wrote the manuscript.

COMPETING FINANCIAL INTERESTS

The authors declare no competing financial interests.

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ONLINE METHODS

Sample collection and processing. Tumor tissues were obtained from patients diagnosed with colorectal cancer, after written informed consent was obtained, from the National Cancer Center of Singapore, Singapore General Hospital. Our study was approved by the Singhealth Institutional Review Board: 2013/110/B. Patients (n = 11) diagnosed with stage 2–4 CRC were selected for this study. Primary tumors were surgically removed and kept in sterile and cold RPMI medium until processing. Normal Mucosa (NM) was sampled far from neoplastic tissue. Fresh resected tumor and normal mucosa samples were processed as described below. Briefly, samples were minced on a plate and enzymatically digested with collagenase IV (Gibco) and DNase I (Worthington) for 30 min at 37 °C, with agitation. After digestion, samples were sieved through a 100-μm cell strainer, washed with 1% BSA and 2 mM EDTA in PBS, and centrifuged for 8 min at 500*g*. The previous step was repeated using a 40-μm cell strainer. Pelleted cells were then resuspended in 2 mM EDTA in PBS and assessed for viability and size using a Countess instrument (Life Technologies).

Single-cell RNA–seq. Dissociated single cells were stained for viability assessment using Calcein-AM and Ethidium Homodimer 1 (Live/Dead Kit, Life Technologies) before being loaded onto RNA–seq IFC-C1 chips (5–10 and 10–17 μm in size). Optimal buoyancy for dissociated single cells was achieved by adding Cell Suspension Buffer and Cell Wash Buffer before loading onto the IFC chips. Once cell capture was achieved, the IFC chips were imaged using a Nikon microscope. On the basis of bright-field and Calcein-AM imaging, only single cells were selected for subsequent library preparation and sequencing analysis. scRNA–seq was performed according to the manufacturer's instructions on C1 machines using the SMARTer Ultra-Low RNA Kit for Illumina (Clontech, 634948). Briefly, after cell lysis, mRNA was isolated and converted into cDNA using modified oligo(dT) primers. Pre-amplification of cDNA was performed using the Advantage 2 PCR kit (Clontech, 639206) for 21 cycles (C1 protocol). cDNAs were subsequently collected for each single cell, individually quantified using the Quant-iT PicoGreen dsDNA Assay kit (Life Technologies) and checked for quality using High-Sensitivity DNA Chips on a Bioanalyzer instrument (Agilent Technologies). The same protocol was used to perform scRNA–seq on the following human cell lines: A549 (lung carcinoma; ATCC), GM12878 (lymphoblastoid; Coriell Institute), H1437 (lung adenocarcinoma; ATCC), IMR90 (fetal lung fibroblast; ATCC), H1 (human embryonic stem cell; Wicell[53]) and K562 (myelogenous leukemia; ATCC). H1 cells were systematically tested for mycoplasma contamination. For GM12878 and H1 cells, two independent experiments were conducted to test for batch effects.

Library preparation and sequencing. DNA libraries were prepared for sequencing after discarding low-quality single cells (on the basis of cDNA concentration and quality). We used the Nextera XT DNA Sample Preparation kit in combination with the Nextera XT DNA Sample Preparation Index kit (96 indices) (Illumina) for individual library preparation, according to the manufacturer's instructions. We subsequently pooled up to 96 libraries for clean-up using Agencourt AMPure XP beads (Agencourt Bioscience). Pooled and multiplexed libraries were sequenced using the 101-bp paired-end protocol on the HiSeq 2000 platform (Illumina). We performed scRNA–seq on a total of 1,591 cells isolated from CRC tumors and matched normal mucosa and also on 630 cells from cell lines.

Sequence data preprocessing and quality control. Raw FASTQ files were aligned to the hg19 human genome assembly using TopHat 2.1.0 and GENCODEv19 gene models. The TopHat parameters were tuned for scRNA–seq: --read-edit-dist was set to 3 and --read-realign-edit-dist was set to 0. The max-multihits parameter was set to 1 so that only uniquely mapped reads were included in the alignment BAM file. Gene expression was quantified as FPKM using Cuffdiff-2.2.1 with the --frag-bias-correct parameter enabled and --library-norm-method set to classic-fpkm. To minimize the systematic scaling effect of short RNAs on transcriptome-wide FPKM values, we masked rRNAs, tRNAs, snRNAs and snoRNAs using the --mask-file option of Cuffdiff. Raw read count values were also exported by Cuffdiff for quality control analysis.

Quality control analysis. FASTQ reads (FASTQR) were calculated as the number of paired-end sequences in the FASTQ file. Mapped reads in BAM files (BAMMR) were defined as the number of uniquely mapped reads in the aligned BAM file. Exonic reads (ER) were calculated by summing raw read counts across all genes. The rate of exonic reads (ROER) represents the ratio between ER and BAMMR. The number of detected genes (NODG) was calculated as the total number of genes with FPKM ≥1. The following housekeeping genes were used for quality control criteria analysis: *TFRC*, *ACTB*, *RPLP0*, *PGK1*, *GAPDH*, *LDHA*, *NONO*, *B2M*, *GUSB* and *PPIH*. Cells with NODG ≥1,000, ROER ≥5% and ER ≥0.1 million were selected for downstream analysis. After quality control, we retained a total of 626 cells from CRC and matched normal mucosa and 561 cells from cell lines. All results are based on post-quality-control cells.

Immunohistochemistry. IHC experiments were conducted on snap-frozen tissue sections. Briefly, frozen specimens were embedded in Tissue-Tek OCT (VWR) and cut into 10-μm cryosections on slides. Subsequently, samples were fixed with 4% paraformaldehyde for 10 min at room temperature. Blocking was achieved using 10% serum, 1% BSA (Sigma), 0.3 M glycine and 0.1% Tween-20 in PBS for 1 h at room temperature. Tissues were stained at 4 °C overnight using primary antibodies diluted in blocking solution: anti-TGF-β (eBioscience, 16924385; 1:50), anti-CD90, anti-E-cadherin (Cell Signaling Technology, 3195; 1:200), anti-α-SMA (Abcam, ab7817; 1:400), anti-MMP2 (Abcam, ab37150; 1:200), anti-PDGFA (Santa Cruz Biotechnology, sc-9974; 1:400), anti-LUM (Abcam, ab168384; 1:50), anti-VIM (R&D, AF2105; 1:20), anti-TAGLN (Sigma, HPA019467; 1:50) and anti-FAP (Santa Cruz Biotechnology, sc-65398; 1:1,000). After washing with PBS, the samples were incubated with the corresponding labeled secondary antibodies for 1 h at room temperature in PBS: anti-mouse AF594 (A11032), anti-rabbit AF594 (R37119), anti-rabbit AF647 (A31573) and anti-rabbit AF405 (A31556) (Invitrogen; 1:500 for all). CD90 was detected using AF647- or FITC-conjugated anti-human-CD90 antibody (BioLegend, 328107 and 328116; 1:50). PDPN was detected using AF488-conjugated primary antibody (BioLegend, 337005; 1:50). Nuclear counterstaining was performed using TO-PRO or DAPI (Invitrogen). Slides were mounted with ProLong Gold Antifade Mountant (Life Technologies). Tissues were visualized by confocal imaging using Zeiss LSM510 inverted microscopy. Images were processed with ImageJ software.

Single-molecule FISH. Before sectioning, tissues were fixed in cold 4% paraformaldehyde (Electron Microscopy Sciences, 15714) for at least 16 h at 4 °C. After fixation, the tissue samples were washed in cold 1× PBS and embedded in Tissue-Tek OCT compound (Sakura FineTek, 4583). Using a Leica CM3050 S cryostat, the samples were sectioned to 7-μm thickness on 22 × 22 mm No. 1 coverslips (Deckglässer) coated with poly(L-lysine) (Sigma, P4707-50 ml). The tissue sections were then washed in cold 1× PBS before post-fixation in 4% paraformaldehyde for 15 min at room temperature. After fixation, the tissue samples were treated with 0.1% sodium borohydride (Sigma, 480886) for 5 min at room temperature to reduce background autofluorescence. The samples were washed twice with 1× PBS before incubation in 0.5% Triton X-100 in 1× PBS for 1 min at room temperature to permeabilize the tissue. The samples were then washed twice in 1× PBS, and a 10% formamide wash buffer, containing 10% deionized formamide (Ambion, AM9344) and 2× saline sodium citrate (SSC), was added. The samples were incubated in the 10% formamide wash buffer at 37 °C for 5 min before proceeding to the hybridization step.

Labeling of FISH probes. Oligonucleotides carrying an amine group were purchased from Integrated DNA Technologies, presuspended in Tris-EDTA buffer. The oligonucleotides corresponding to each gene were pooled together to a final concentration of 200 μM. The combined oligonucleotides were labeled with Cy3 and Cy5 monofunctional reactive dyes (Amersham, Cy3-PA23001 and Cy5-PA25001 MonoReactive Dye Packs). First, the dyes were reconstituted with DMSO and mixed with the pooled oligonucleotides in pH 8.5 sodium borate buffer with the NH_3:dye ratio at 1:20. The mixtures were incubated on a shaker at room temperature for at least 5 h in the dark. Then, the oligonucleotides were precipitated by adding 1/10 of the reaction solution volume of 3 M sodium acetate and 2.5 volumes of cold reaction solution (prechilled at −20 °C) to the reaction solution and incubating at −20 °C for 60 min. Next, the solution was centrifuged for 30 min to recover the DNA.

The pellet was washed three times with 70% ethanol before resuspension in 20 µl of nuclease-free water. The labeled FISH probes were further purified through HPLC (Agilent 1260 Infinity) using the Phenomenex Clarity C18 column and stored at −20 °C.

The FISH probes were diluted to a final concentration of 400 nM in FISH hybridization buffer (10% deionized formamide, 2× SSC, 20 mg/ml *Escherichia coli* tRNA (Life Technologies, 15401-11), 2 mM vanadyl ribonucleoside complex (New England BioLabs, S1402S) and 10% dextran sulfate (Sigma, D8906-100G)). The hybridization buffer was added to the samples, and the samples were incubated in a humidified chamber at 37 °C for 4 h. Following hybridization, the samples were washed twice in 10% formamide wash buffer, incubating at 37 °C for 15 min per wash. The samples were then washed twice with 2× SSC and stained with Hoechst H33342 (Sigma, B2261). The samples were either imaged immediately or were stored for no longer than 24 h at 4 °C in 2× SSC before imaging. The imaging buffer consisted of 2× SSC, 50 mM Tris-HCl pH 8, 10% glucose, 2 mM Trolox (Sigma, 238813), 0.5 mg/ml glucose oxidase (Sigma, G2133) and 40 µg/ml catalase (Sigma, C30).

Imaging. A custom-built microscope was constructed around the Nikon Ti-E body, MS-200 ASI X-Y stage, 1.45 N.A. CFI Plan Apo Lambda 100× oil-immersion objective and Andor iXon Ultra 888 EMCCD camera. Hoechst, Cy3 and Cy5 were excited by 405-nm (LuxX, 405-20), 561-nm (DPSS, 561-50) and 638-nm (LuxX, 638-100) solid-state lasers (Omicron), respectively. Emission was filtered with a custom dichroic ZT405/488/561/640/730rpc-uf (Chroma) and emission filters ET455/50M, ET595/50M and ET685/50 (Chroma), respectively. The samples were imaged using wide-field imaging with oblique-incidence illumination controlled with the Nikon motorized laser TIRF illuminator. Images were acquired with 0 gain, and the exposure time was 1 s.

RNAscope. RNA detection was conducted on snap-frozen tissue cryosections. Briefly, frozen specimens were embedded in Tissue-Tek OCT (VWR) and sectioned at 20 µm on slides. Samples were fixed and processed according to the manufacturer's instructions for fresh-frozen tissue. The samples were incubated with RNAscope Probe-Hs-REG4 (312071) and RNAscope Probe-Hs-MUC2 (312871) probes and visualized using the RNAscope Multiplex Fluorescent Assay. On the same slides, E-cadherin was subsequently detected by IHC, as described above, using the anti-E-cadherin primary antibody (Cell Signaling Technology, 3195; 1:200) and the Alexa Fluor 647–conjugated donkey anti-mouse secondary antibody (Invitrogen, A31571; 1:500). Nuclear counterstaining was performed using DAPI, and slides were mounted with ProLong Gold Antifade mounting medium (Life Technologies). Images were acquired on the Zeiss LSM510 inverted confocal microscope and processed using ImageJ software.

Algorithm evaluation on the benchmark cell line scRNA–seq data set. We used eight existing methods to cluster cell line scRNA–seq profiles: All-HC, HiLoadG-HC, BackSPIN, RaceID2, Seurat, VarG-HC, VarG-PCAproj-HC and VarG-tSNEproj-HC. For All-HC, genes with FPKM values $\geq 1 \times 10^{-3}$ in at least two cells were selected as features and $\log_{10}$ (FPKM) values with pseudocounts of 1×10^{-3} were used for hierarchical clustering with 'average'-linkage neighbor joining. For HiLoadG-HC, genes with loading ranks in the top or bottom 100 for PC1, PC2 or PC3 were selected as features and pseudocounted $\log_{10}$ (FPKM) values were used for hierarchical clustering with the average linkage option. For BackSPIN, cells were clustered using the following parameter settings: -f 2000 -v -d 4; this yielded more accurate clustering than the default parameter settings. For RaceID2, default parameters were used to cluster the raw FPKM data without downsampling (other parameter settings did not yield a noticeable improvement). We used $k = 7$ on the basis of the within-cluster-dispersion criterion of RaceID2 as the number of clusters for k-medoids clustering. For Seurat, an online command list provided by the authors of Seurat was followed, with the latent.vars parameter for the RegressOut function set to nUMI and the model.use parameter for the RegressOut function set to linear. An alternative test with model.use set to negbinom showed similar performance (data not shown). For VarG-HC, the top 1,000 most highly variable genes were selected using BackSPIN's feature selection criteria, followed by hierarchical clustering on feature genes, quantified as $\log_{10}$ (FPKM) values. For VarG-PCAproj-HC, the approach was the same as for VarG-HC except that PCA was used for dimension reduction after feature selection, such that 90% of the variance in the expression matrix was retained. For VarG-tSNEproj-HC, the approach was the same as for VarG-HC except that a 2D tSNE projection was used for dimension reduction after feature selection.

Algorithm evaluation on the melanoma scRNA–seq data set. Expression data for 4,645 melanoma tumor cells were downloaded from the Gene Expression Omnibus (GEO) via accession GSE72056. We focused on the six major cell types in the data set and filtered cells on the basis of the list of marker genes used in the original study[6]: T cells (*CD2*, *CD3D*, *CD3E*, *CD3G*), B cells (*CD19*, *CD79A*, *CD79B*, *BLK*), macrophages (*CD163*, *CD14*, *CSF1R*), endothelial cells (*PECAM1*, *VWF*, *CDH5*), CAFs (*FAP*, *THY1*, *DCN*, *COL1A1*, *COL1A2*, *COL6A1*, *COL6A2*, *COL6A3*) and malignant melanoma cells (*MIA*, *TYR*, *SLC45A2*). We excluded cells in which none of the markers were detected, and the rest were hierarchically clustered on the basis of marker gene expression to assign a total of 4,057 cells to the six cell types.

RCA: global panel. Raw reference bulk transcriptomes from HumanU133A/GNF1H Gene Atlas and the Primary Cell Atlas were downloaded from BioGPS. Quantile normalization was applied separately to the Human U133A/GNF1H Gene Atlas and the Primary Cell Atlas. Feature genes were selected on the basis of sample-specific expression in either data set. For GNF1H data, a gene was included in the feature set if the $\log_{10}$ (fold change) of its expression in any sample relative to the median across all samples exceeded 1. The same procedure was used to define the feature gene set for the Primary Cell Atlas reference panel except that the threshold was changed from 1 to 1.1 to bring the number of feature genes in line with that for the GNF1H panel. A total of 4,717 genes were selected as features for GNF1H and 5,209 genes were selected for the Primary Cell Atlas.

The FPKM profile of each single cell was projected onto each sample in the GNF1H panel by calculating the Pearson correlation coefficient between the $\log_{10}$ (FPKM) vector from scRNA–seq (pseudocount = 1) and the $\log_{10}$ (expression) vector from the reference bulk transcriptome. Only GNF1H feature genes were used in calculating the correlation coefficient. Projections onto Primary Cell Atlas transcriptomes were performed in the same manner, using the corresponding feature genes. For each single cell, all of the above-described correlation coefficients were raised to the fourth power and then concatenated to form a vector, which was then z score transformed to define the projection vector for the cell.

Single cells were clustered using average-linkage hierarchical clustering of their projection vectors. To identify cell type clusters within the resulting dendrogram, we used the cutreeDynamic function from the WGCNA package, with deepSplit = 1 and mingroupSize = 5.

RCA: self-projection. Single-cell raw FPKM values were normalized using pseudocounted quantile (pQ) normalization[54] and then $\log_{10}$ transformed (pseudocount = 1). Feature selection was performed to prioritize genes whose expression vectors possessed the on–off characteristics of highly specific cell type markers. In other words, we selected genes that were highly expressed in at least one of every 15 cells and were also undetected in at least one of every 15 cells. A gene was said to be highly expressed in any given cell if its expression was greater than $\log_{10}$ (50) and also greater than its own median non-pseudocounted expression value across all cells.

The FPKM profile of each single cell was projected onto the entire set of cells by calculating the Pearson correlation coefficient between the respective $\log_{10}$ (FPKM) vectors (pseudocount = 1). Only single-cell feature genes were used in calculating the correlation coefficient. The vector of correlation coefficients for each cell was z score transformed to define its projection vector. Single cells were clustered using average-linkage hierarchical clustering of their projection vectors. To identify cell type clusters within the resulting dendrogram, we used the cutreeDynamic function from the WGCNA package, with deepSplit = 3 and min group Size = 5.

Constructing the colon epithelium panel. The colon epithelium panel was constructed by simulating the bulk transcriptome of each of the nine normal mucosa epithelial clusters. Bulk transcriptomes were created by averaging the pQ-normalized and log-transformed expression vectors of all single cells within the corresponding cell type cluster. A gene was included in the feature set for clustering if the $\log_{10}$ (fold change) of its 'bulk' expression in any cell cluster relative to the median across all nine cell clusters exceeded 1.5.

To assess the correspondence of the identified normal mucosa epithelial subtypes (**Fig. 3a**) to known intestinal cell types, we examined the expression of known markers of enterocyte, goblet, stem/TA, enteroendocrine, Paneth and tuft cells. We found high expression of enterocyte markers and also markers of goblet and stem/TA cells, but markers of the other three cell types were only minimally detected. For each detected cell type, we averaged the expression values of matching marker gene sets (enterocyte type 1, enterocyte type 2, goblet, ribosomal, bottom crypt) to obtain metagene expression scores. Thus, the expression vector of each cell was projected onto a five-dimensional metagene space. We then used the Wilcoxon rank-sum test to detect cell-type-specific differential expression by comparing cells from each cluster to all other cells. Metagenes with $P \leq 0.05$ and fold change ≥ 2 were labeled as upregulated, while those with $P \leq 0.05$ and fold change ≤ 0.5 were labeled as downregulated.

Normalization and differential expression analysis. scRNA–seq data were pQ normalized[54], and differential expression analysis was performed using NODES[54]. Genes with a differential expression q value ≤ 0.05 and fold change ≥ 2 were selected as differentially expressed genes. Bulk RNA–seq data were quantile normalized, and the Wilcoxon rank-sum test was used to identify differentially expressed genes between normal and tumor samples. Genes with Benjamini–Hochberg FDR $\leq 1 \times 10^{-10}$ and fold change ≥ 4 were defined as bulk differentially expressed genes.

Bulk data projection and patient survival analysis. The six most abundant tumor cell groups were selected for patient survival analysis: stem/TA-like cells, other epithelial cells, fibroblasts, T cells, B cells and myeloid cells. The mean expression profile of each cell group was calculated by averaging log-transformed single-cell FPKM values. The six mean expression profiles were then quantile normalized. Expressed genes were defined as those whose expression exceeded 1 FPKM in at least one cell group. Feature genes were then defined as expressed genes whose expression in at least one cell group exceeded the median across all six groups by at least a factor of 4.

Bulk-transcriptomic data were collected from TCGA, GSE14333, the PRECOG database[37], and GSE33113, GSE37892 and GSE39582 used in a recent study[38] and log transformed if not already in the log domain. Bulk transcriptomes were then projected onto the above-described cell group feature matrix via Pearson correlation. As before, correlation coefficient vectors were z score transformed and then used to cluster the bulk transcriptomes by average-linkage hierarchical clustering. Kaplan–Meier survival curves were generated using the survfit function in R, and the P values of survival differences were calculated using the survdiff function in R.

Data availability. scRNA–seq data have been deposited in NCBI GEO with accession GSE81861. Sequencing data have been deposited at the European Genome-phenome Archive (EGA), which is hosted by the EBI and CRG, under accession EGAS00001001945.

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nature genetics

AMENDMENTS

https://doi.org/10.1038/s41588-018-0299-1

Author Correction: Reference component analysis of single-cell transcriptomes elucidates cellular heterogeneity in human colorectal tumors

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Correction to: *Nature Genetics* https://doi.org/10.1038/ng.3818, published online 20 March 2017.

In the version of the article published, the author list is not accurate. Igor Cima and Min-Han Tan should have been authors, appearing after Mark Wong in the author list, while Paul Jongjoon Choi should not have been listed as an author. Igor Cima and Min-Han Tan both have the affiliation Institute of Bioengineering and Nanotechnology, Singapore, Singapore, and their contributions should have been noted in the Author Contributions section as "I.C. preprocessed Primary Cell Atlas data with inputs from M.-H.T." The following description of the contribution of Paul Jongjoon Choi should not have appeared: "P.J.C. supported the smFISH experiments." In the 'RCA: global panel' section of the Online Methods, the following sentence should have appeared as the second sentence, "An expression atlas of human primary cells (the Primary Cell Atlas) was preprocessed similarly to in ref. 55," with new reference 55 (Cima, I. et al. Tumor-derived circulating endothelial cell clusters in colorectal cancer. *Science Transl. Med.* **8**, 345ra89, 2016).

Published online: 12 November 2018
https://doi.org/10.1038/s41588-018-0299-1